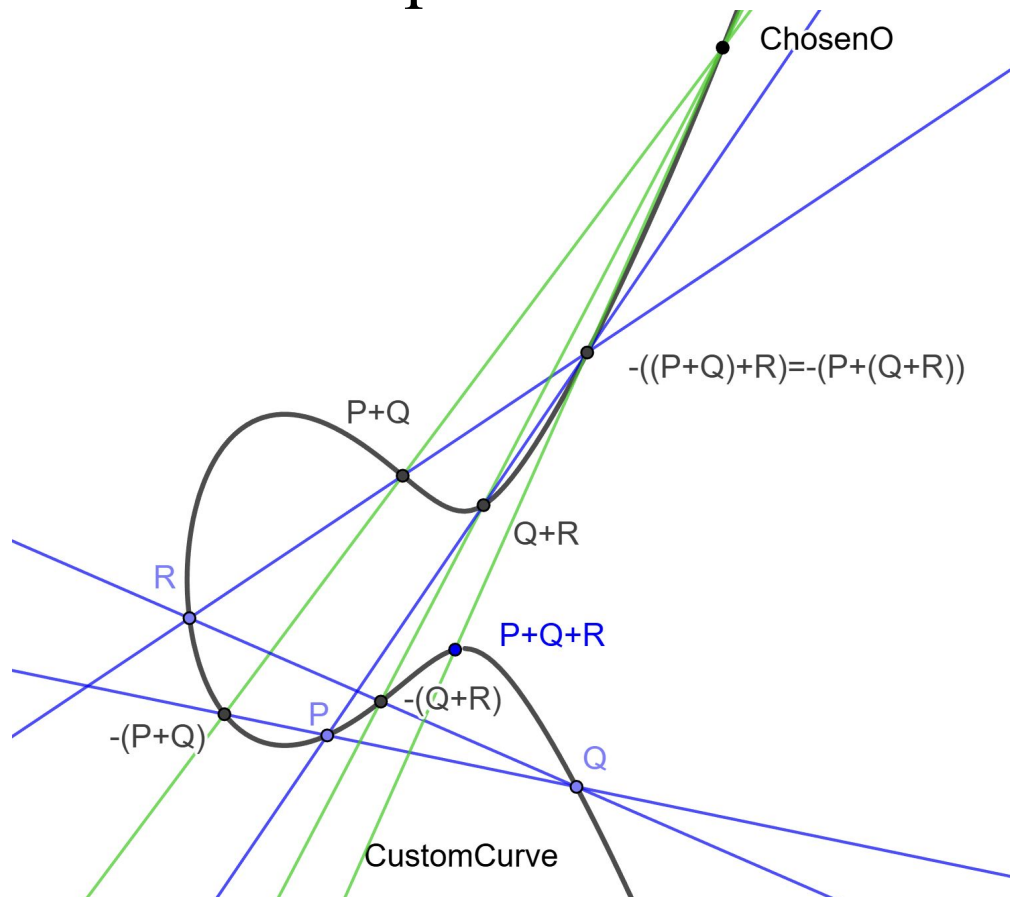


How to do addition in elliptic curve?

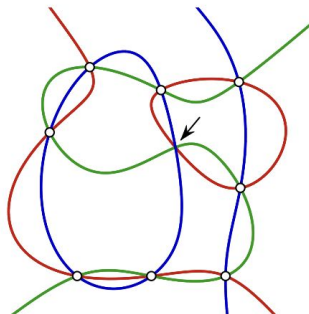


Cayley-Bacharach Formulas

Qingchun Ren, Jürgen Richter-Gebert and Bernd Sturmfels

Abstract

The Cayley-Bacharach Theorem states that all cubic curves through eight given points in the plane also pass through a unique ninth point. We write that point as an explicit rational function in the other eight.



All cubics passing through the eight white points meet in a unique ninth point

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CAYLEY-BACHARACH THEOREMS AND CONJECTURES

DAVID EISENBUD, MARK GREEN, AND JOE HARRIS

of higher degree, such as cubics, became attractive. In any case, Chasles's result is both simpler and more powerful than Pascal's. It is the result that commonly (though incorrectly) goes under the name Cayley-Bacharach.

Theorem CB3 (Chasles). *Let $X_1, X_2 \subset \mathbb{P}^2$ be cubic plane curves meeting in nine points $p_1, \dots, p_9$. If $X \subset \mathbb{P}^2$ is any cubic containing $p_1, \dots, p_8$, then X contains p_9 as well.*

295.

ON THE CONSTRUCTION OF THE NINTH POINT OF INTERSECTION OF THE CUBICS WHICH PASS THROUGH EIGHT GIVEN POINTS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. v. (1862), pp. 222—233.]

I REPRODUCE with additional developments the solution which has been given of this interesting problem.

The equation of a given cubic may be written

$$qU - pV = 0,$$

where $U=0$, $V=0$ are any two conics meeting the cubic in the same four points; $p=0$ is the line joining the remaining two points of intersection of the cubic with the conic $U=0$; and $q=0$ is the line joining the remaining two points of intersection of the cubic with the conic $V=0$; the relation between the arbitrary constant factors implicitly contained in the functions qU and pV is assumed to be properly determined.

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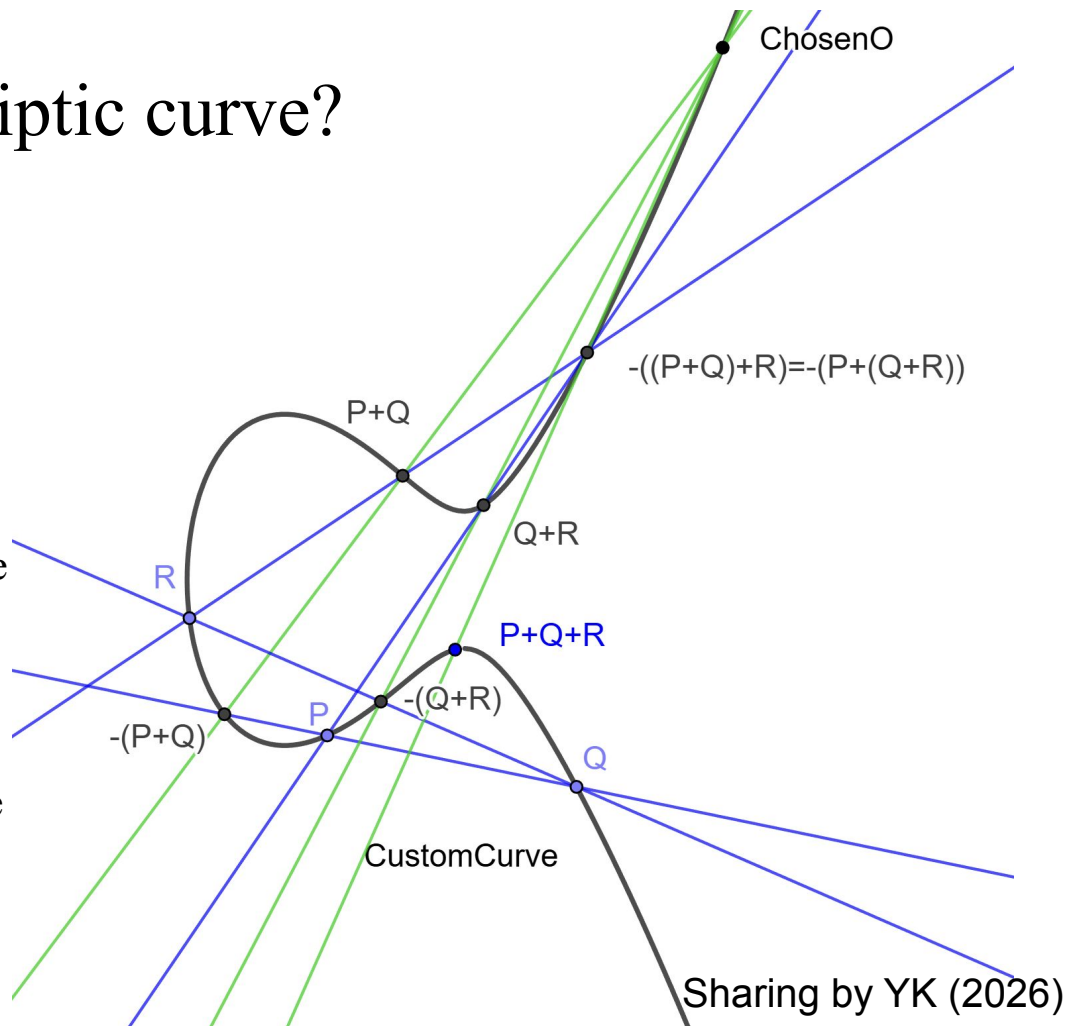
which when eight of the points are given, determine the ninth point. Supposing that the given points are 1, 2, 3, 4, 5, 6, 7, 8, the equations for the determination of the remaining point 9 may be taken to be

$$9 \ (5, 6, 7, 8) = 1234 \ (5, 6, 7, 8),$$

$$9 \ (4, 6, 7, 8) = 1235 \ (4, 6, 7, 8),$$

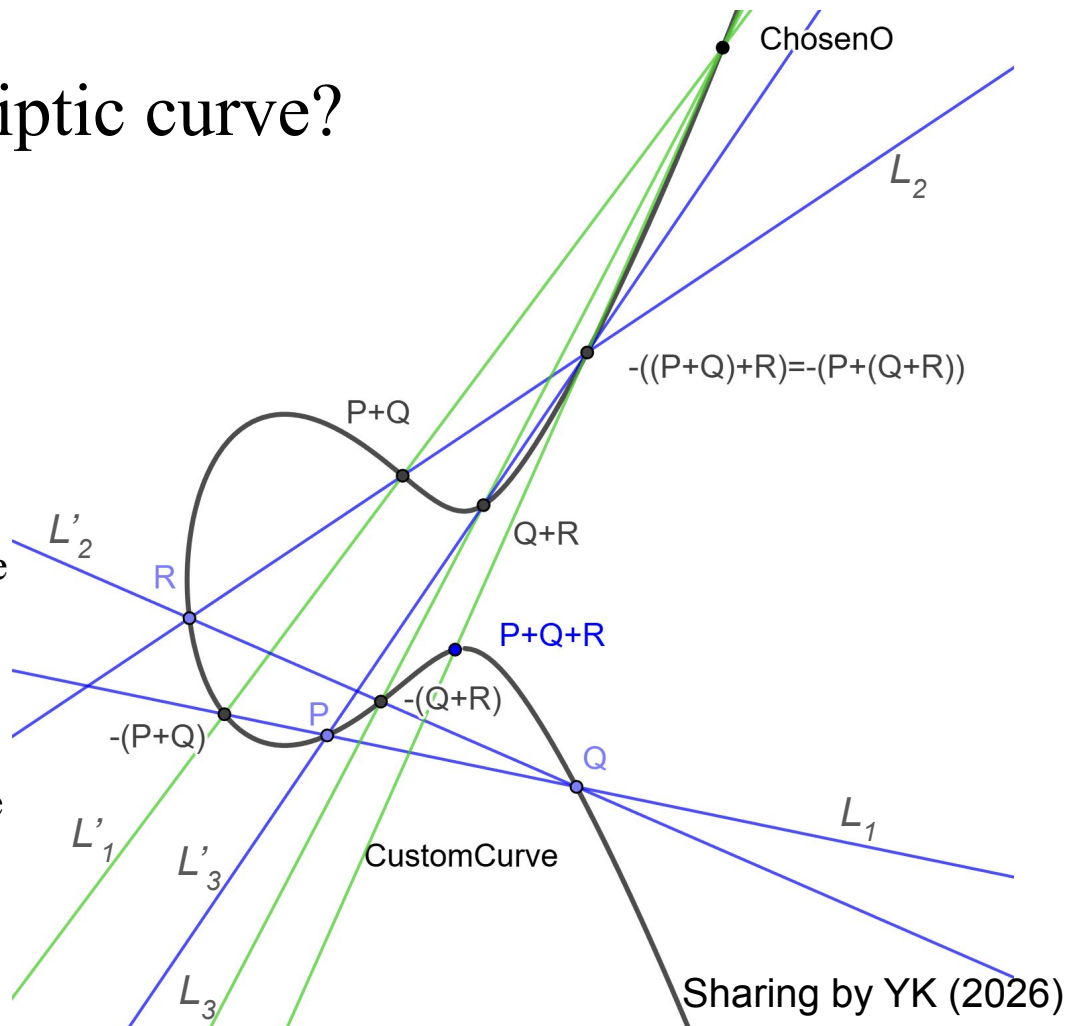
How to do addition in elliptic curve?

1. Start with a cubic curve. Let's say, $C: y^2 = x^3 + ax + b$ with non-zero discriminant, so that a line would intersect the curve at three points over an algebraically closed field with multiplicity considered.
2. Choose a point O on the curve.
3. Given any two points A and B on curve C , the line joining them meets the curve C at a third point, denoted as $-(A+B)$, with multiplicity considered.
4. The line joining O and $-(A+B)$ meets the curve C at a third point with multiplicity considered, that is $A+B$.
5. Cayley-Bacharach Theorem implies the addition defined is associative.



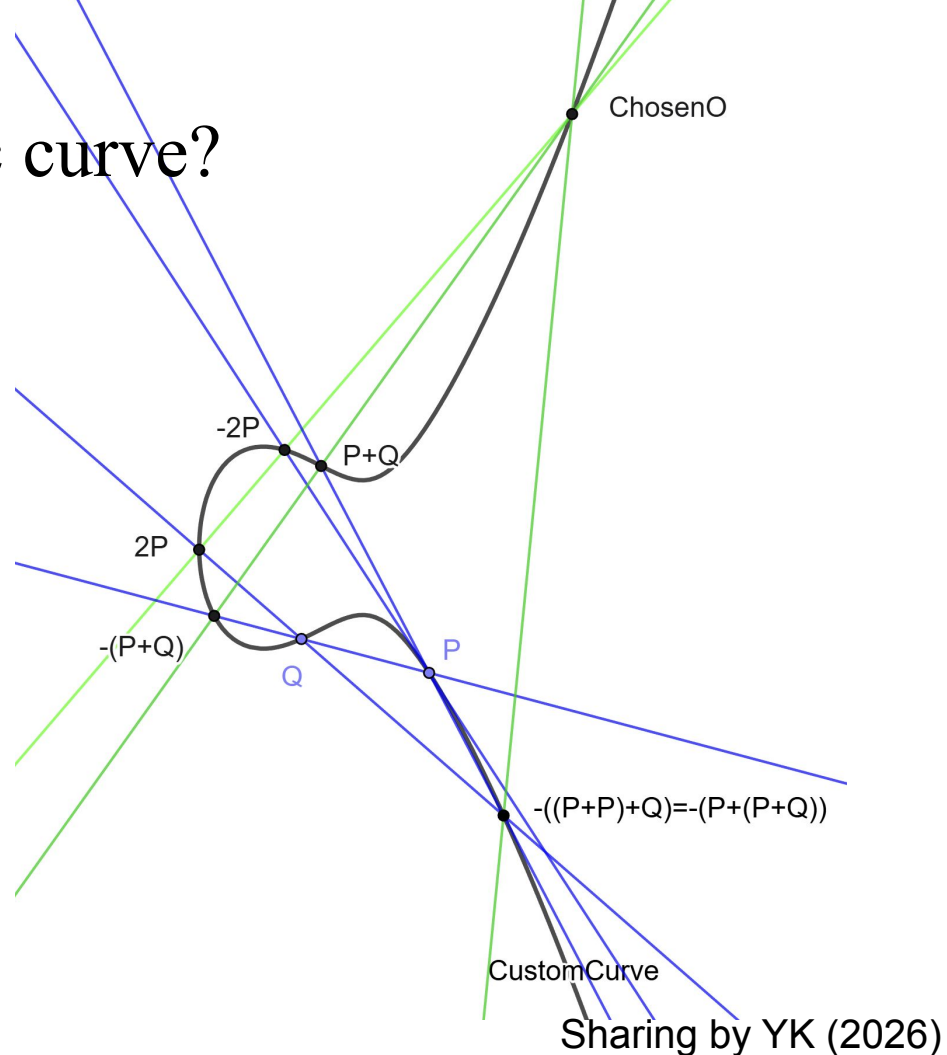
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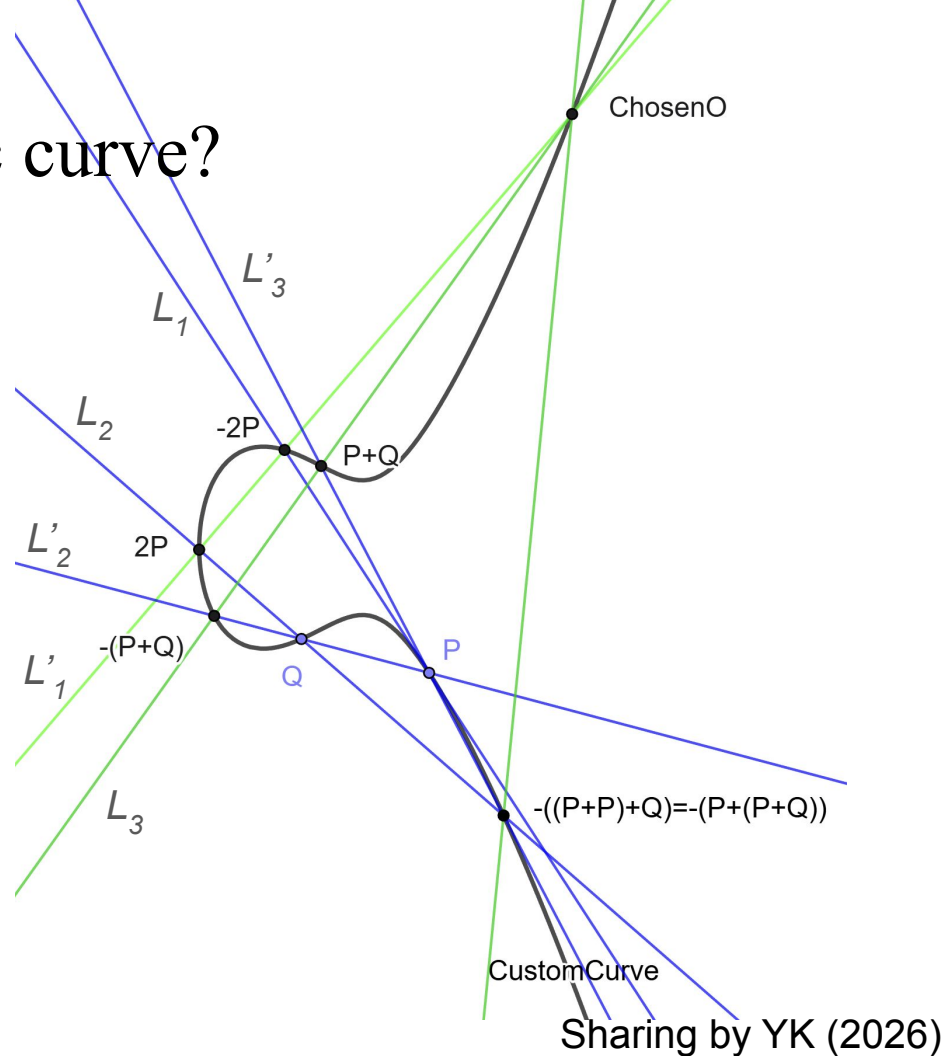
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2. Choose a point O on the curve.
3. When the same point P is given twice, the line tangent to the curve C at P shall meet the curve at a one more point. Such a point is denoted as $-2P$.
4. The line joining O and $-2P$ meets the curve C at a third point with multiplicity considered, that is $2P$.
5. Associativity still holds.



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```

# -----
# 1. 521-bit Mersenne prime & simple curve  $y^2 = x^3 - x + 1$ 
# -----
p = 2**521 - 1
a = p - 1      # a = -1 mod p
b = 1
G = (0, 1)     # base point (0,1) works because  $0^3 - 0 + 1 = 1 = 1^2 \text{ mod } p$ 

print(f"Prime p = 2^521-1 ((p.bit_length()) bits)")
print(f"Curve:  $y^2 = x^3 - x + 1$  over  $F_p$ ")
print(f"Base point G = {G}\n")

# -----
# 2. Elliptic curve arithmetic (standard Weierstrass)
# -----
def point_add(P, Q, a, p):
    if P is None: return Q
    if Q is None: return P
    x1, y1 = P; x2, y2 = Q
    if x1 == x2 and y1 == y2:
        if y1 == 0: return None
        lam = (3 * x1 * x1 + a) * pow(2 * y1, p - 2, p) % p
    else:
        if x1 == x2: return None
        lam = (y2 - y1) * pow(x2 - x1, p - 2, p) % p
    x3 = (lam * lam - x1 - x2) % p
    y3 = (lam * (x1 - x3) - y1) % p
    return (x3, y3)

def scalar_mult(k, P, a, p):
    res = None
    cur = P
    while k:
        if k & 1:
            res = point_add(res, cur, a, p)
        cur = point_add(cur, cur, a, p)
        k >>= 1
    return res

```



```

# --- Bob's long-term key pair ---
print("--- Bob's key pair ---")
mnemonic = generate_mnemonic()
bob_priv = mnemonic_to_private_key(mnemonic)
print(f"Generated Bob's mnemonic: {mnemonic}")

bob_pub = scalar_mult(bob_priv, G, a, p)
print(f"Bob's public key: {bob_pub}\n")

# --- Alice's long-term key pair ---
print("--- Alice's key pair ---")
mnemonic = generate_mnemonic()
alice_priv = mnemonic_to_private_key(mnemonic)
print(f"Generated Alice's mnemonic: {mnemonic}")

alice_pub = scalar_mult(alice_priv, G, a, p)
print(f"Alice's public key: {alice_pub}\n")

```

```

# --- Encryption (Alice encrypts for Bob) ---
print("--- Alice encrypts a message for Bob ---")
message = input("Enter a message to encrypt: ")

# Shared secret = alice_priv * bob_pub
shared_point = scalar_mult(alice_priv, bob_pub, a, p)
shared_x = shared_point[0]
cipher = encrypt(shared_x, message, p)
print(f"Ciphertext (hex): {cipher.hex()}\n")

print("Alice sends (ciphertext, her public key) to Bob.\n")

# --- Decryption (Bob decrypts) ---
print("--- Bob decrypts ---")
print("Bob received Alice's public key and the ciphertext.")

# Bob computes shared secret = bob_priv * alice_pub
recovered_shared_point = scalar_mult(attempt_bob_priv, alice_pub, a, p)
recovered_shared_x = recovered_shared_point[0]

try:
    decrypted = decrypt(recovered_shared_x, cipher, p)
    print(f"Decrypted message: {decrypted}")

```


...

```
=====
ECDH with BIP39 Mnemonic Private Keys
Both parties have long-term key pairs
Curve:  $y^2 = x^3 - x + 1$  over  $F_{(2^{521}-1)}$ 
Encryption: XOR with SHA-256 keystream
=====
```

--- Bob's key pair ---

Generated Bob's mnemonic: soon tone pride because congress laptop photo bleak diagram tape fall october

Bob's public key: (5171454551615612393405042264352694940167908357342320416269299578141204366279931392154698575891277

--- Alice's key pair ---

Generated Alice's mnemonic: fortune move carry chimney cabbage depth case dutch tide main puppy donkey

Alice's public key: (46903309816356045183387982166319477062746998358242959456008027765285711965450205257897567618688

--- Alice encrypts a message for Bob ---

Enter a message to encrypt: Hello

Ciphertext (hex): 2d5260ea0f

Alice sends (ciphertext, her public key) to Bob.

--- Bob decrypts ---

Bob received Alice's public key and the ciphertext.

To decrypt, Bob must provide his private key.

Enter Bob's 12-word BIP39 mnemonic: soon tone pride because congress laptop photo bleak diagram tape fall october

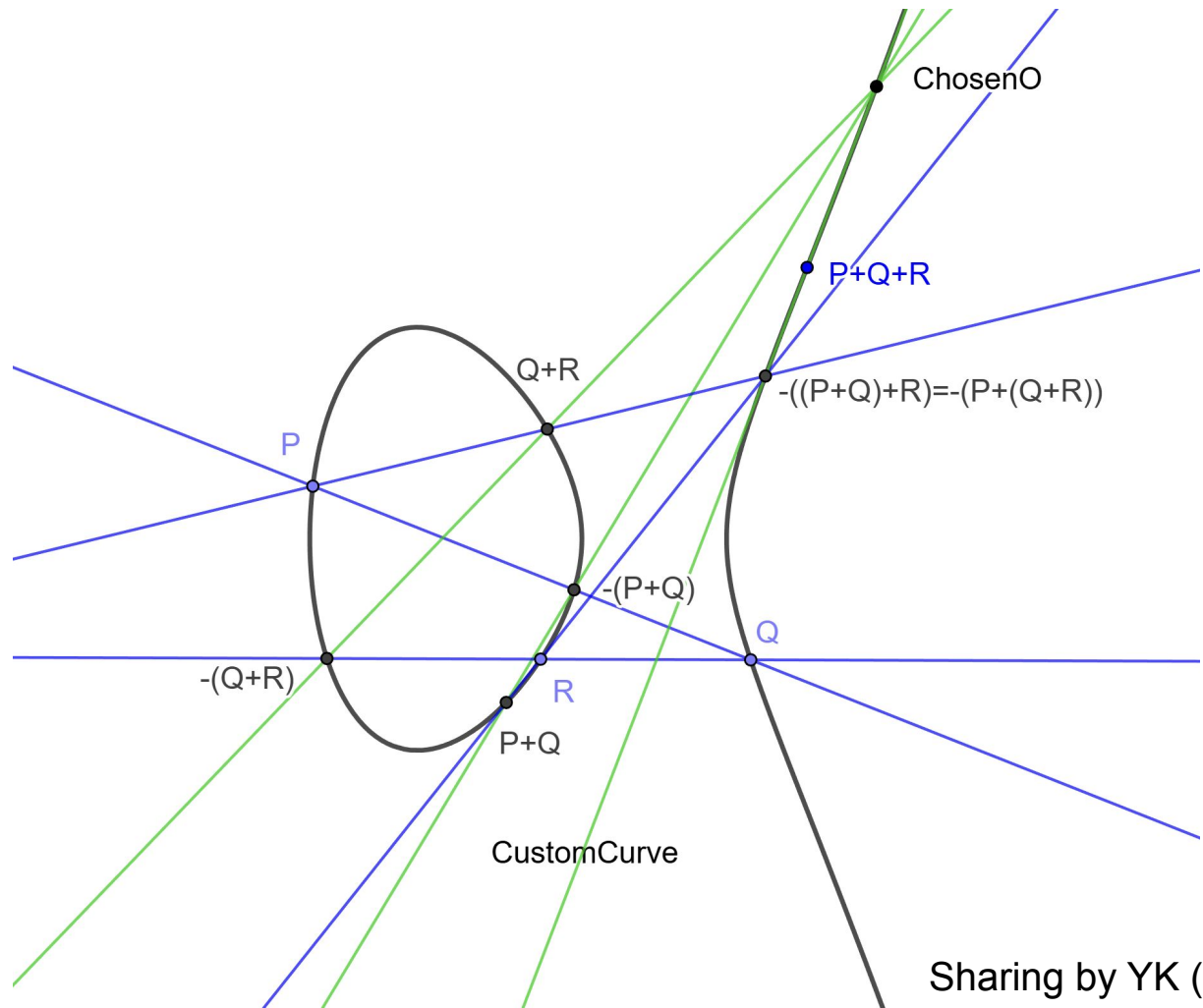


Correct private key!

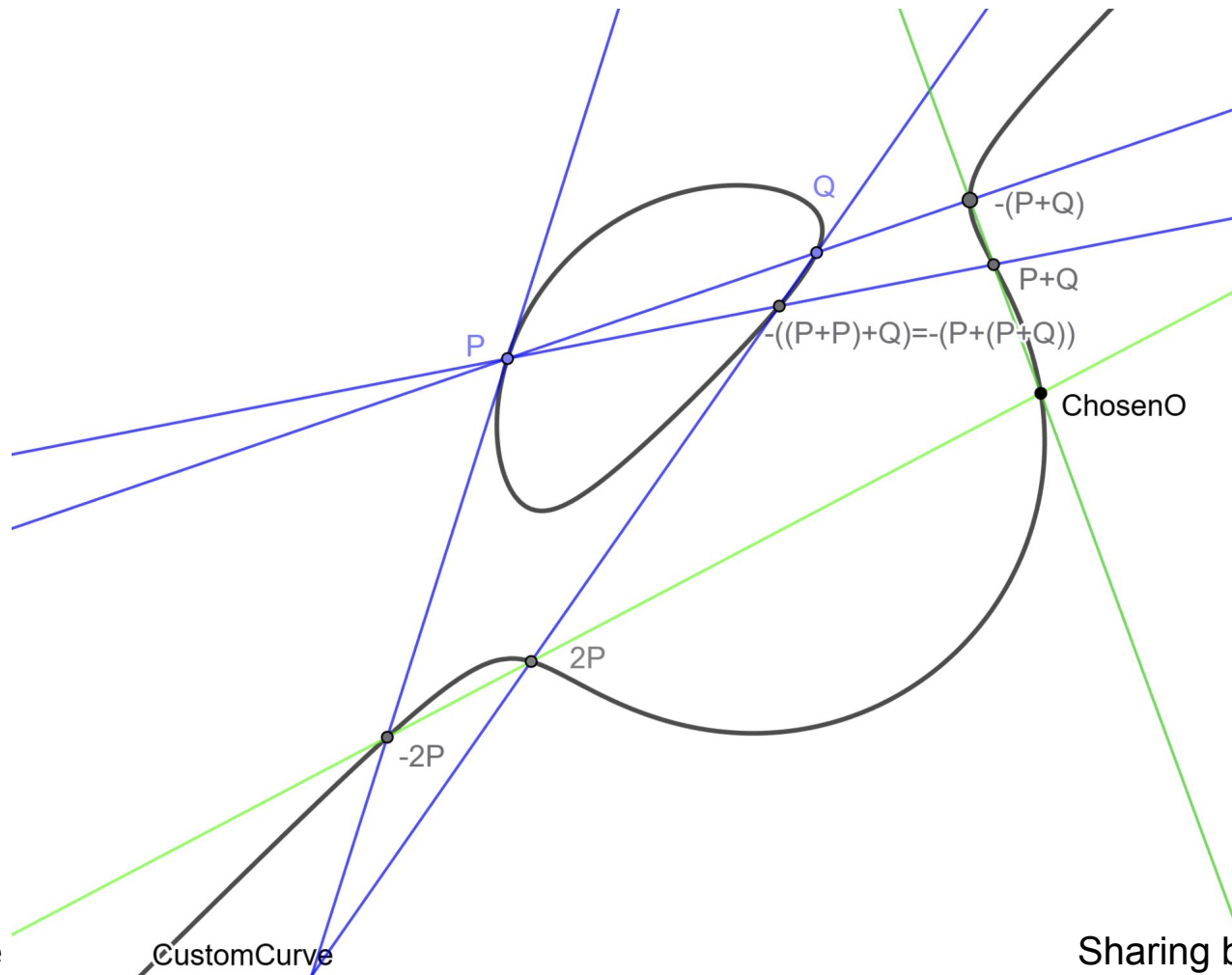
Decrypted message: Hello



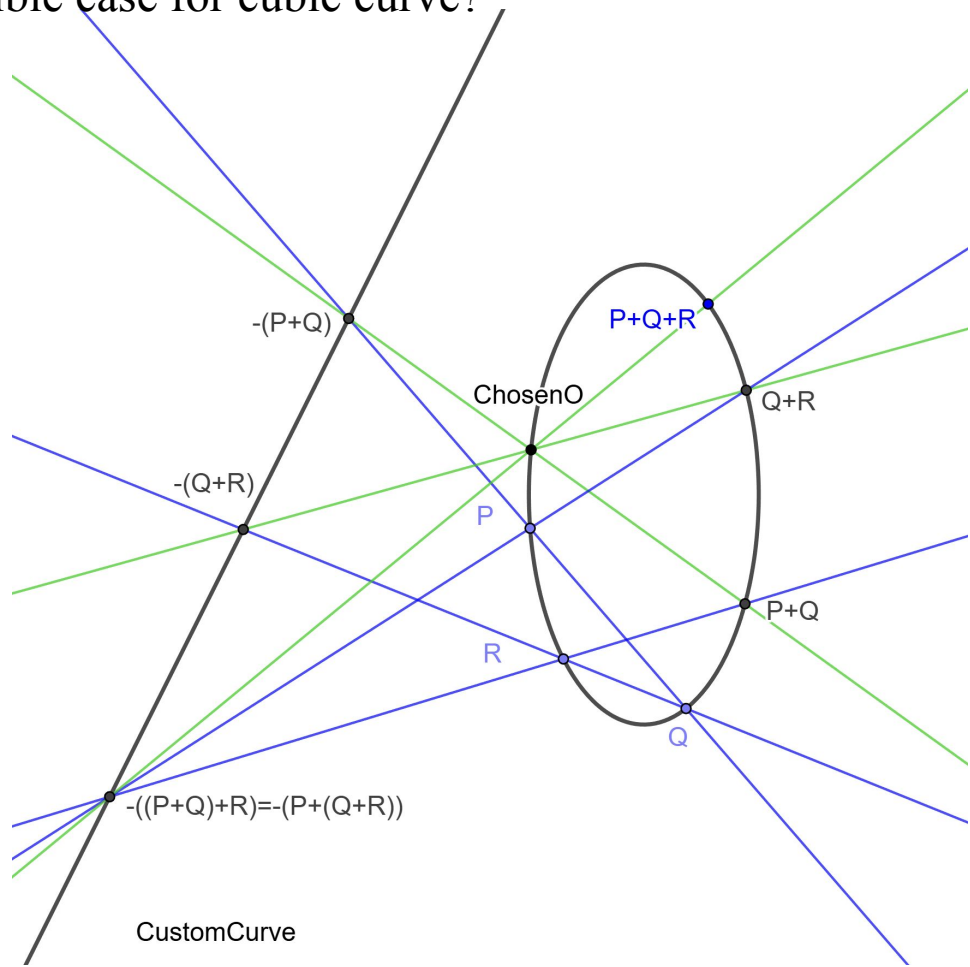
Success - only Bob's private key could decrypt this message.



Sharing by YK (2026)

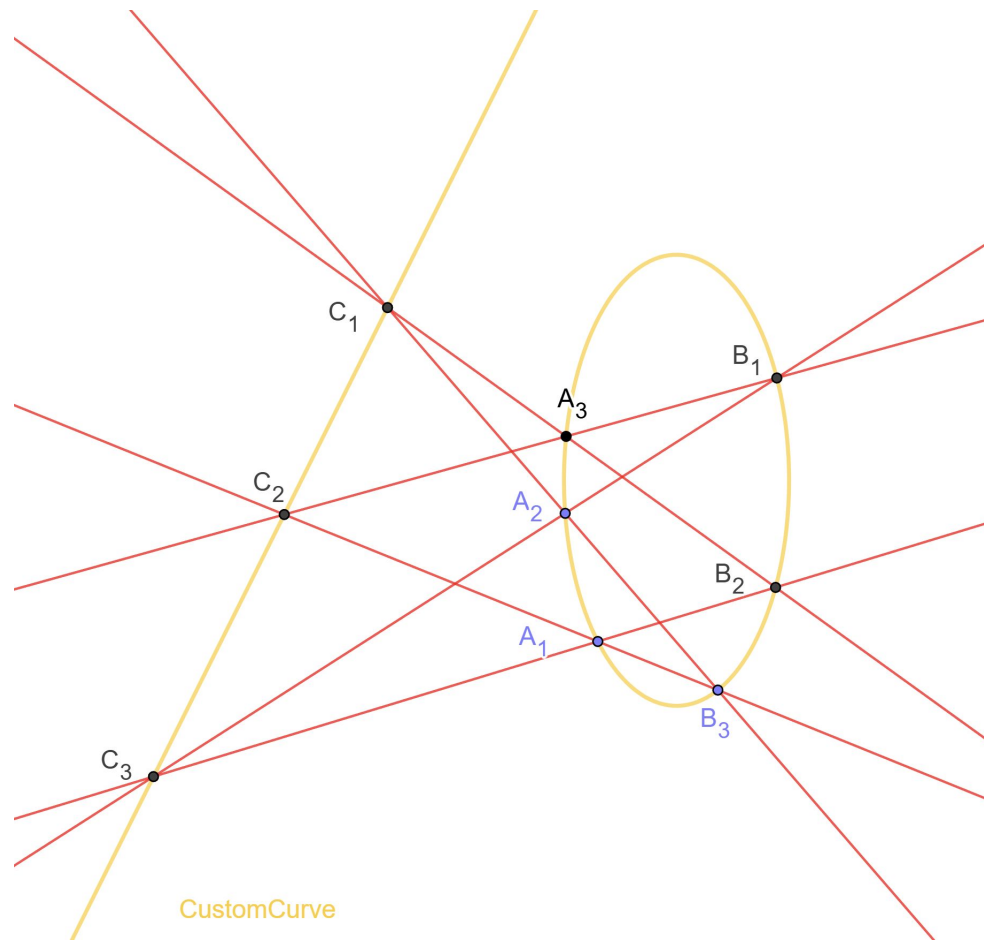


How about some reducible case for cubic curve?

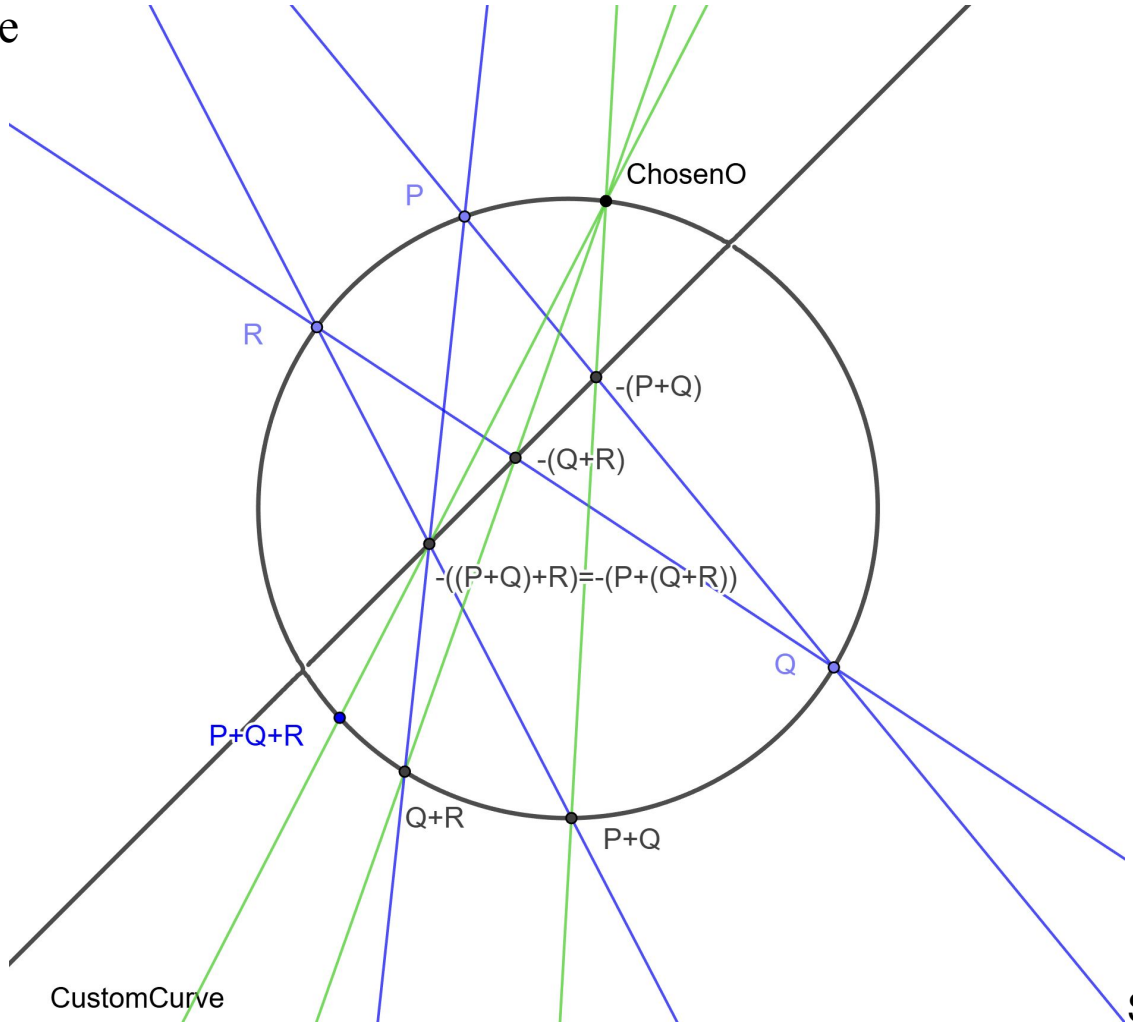


Partial semigroup

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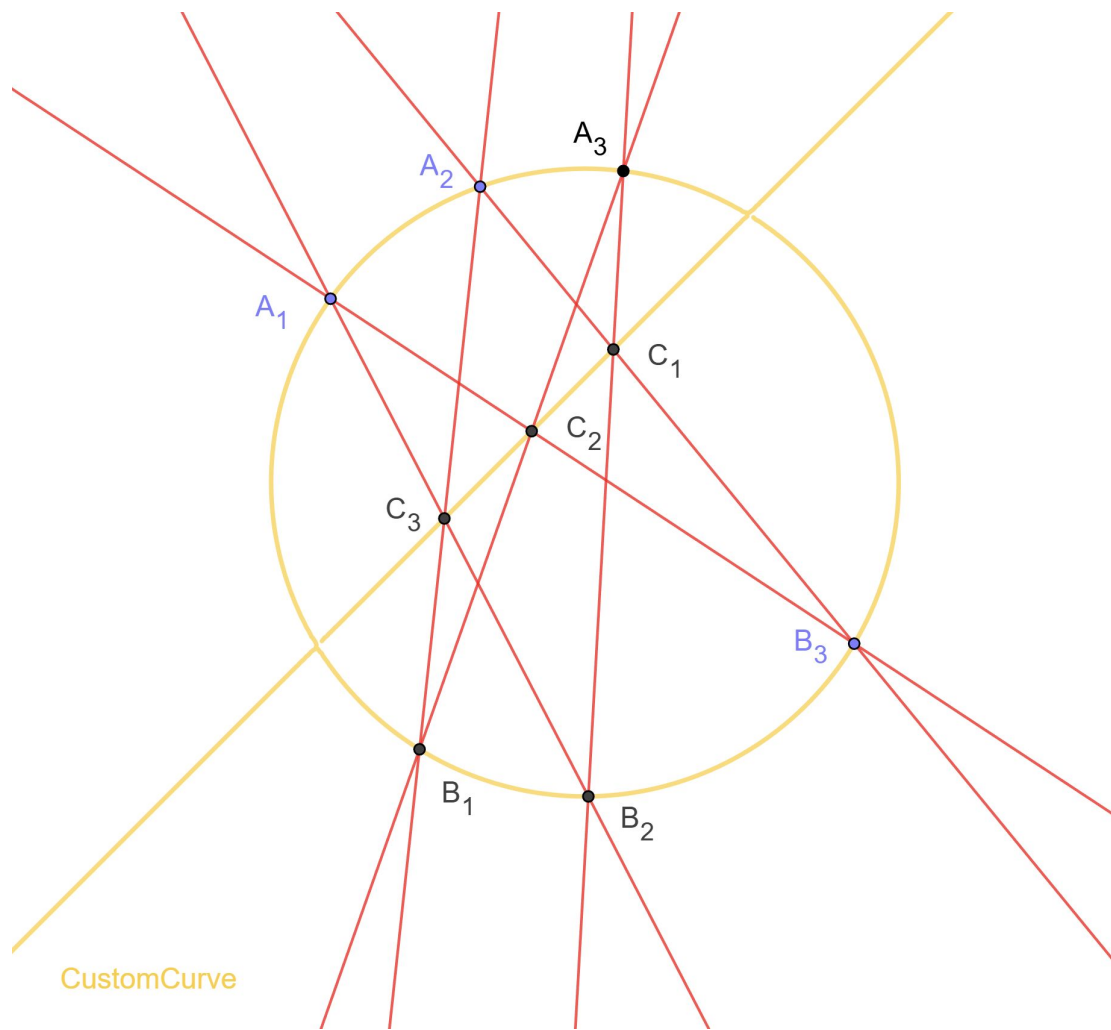
Some reducible case



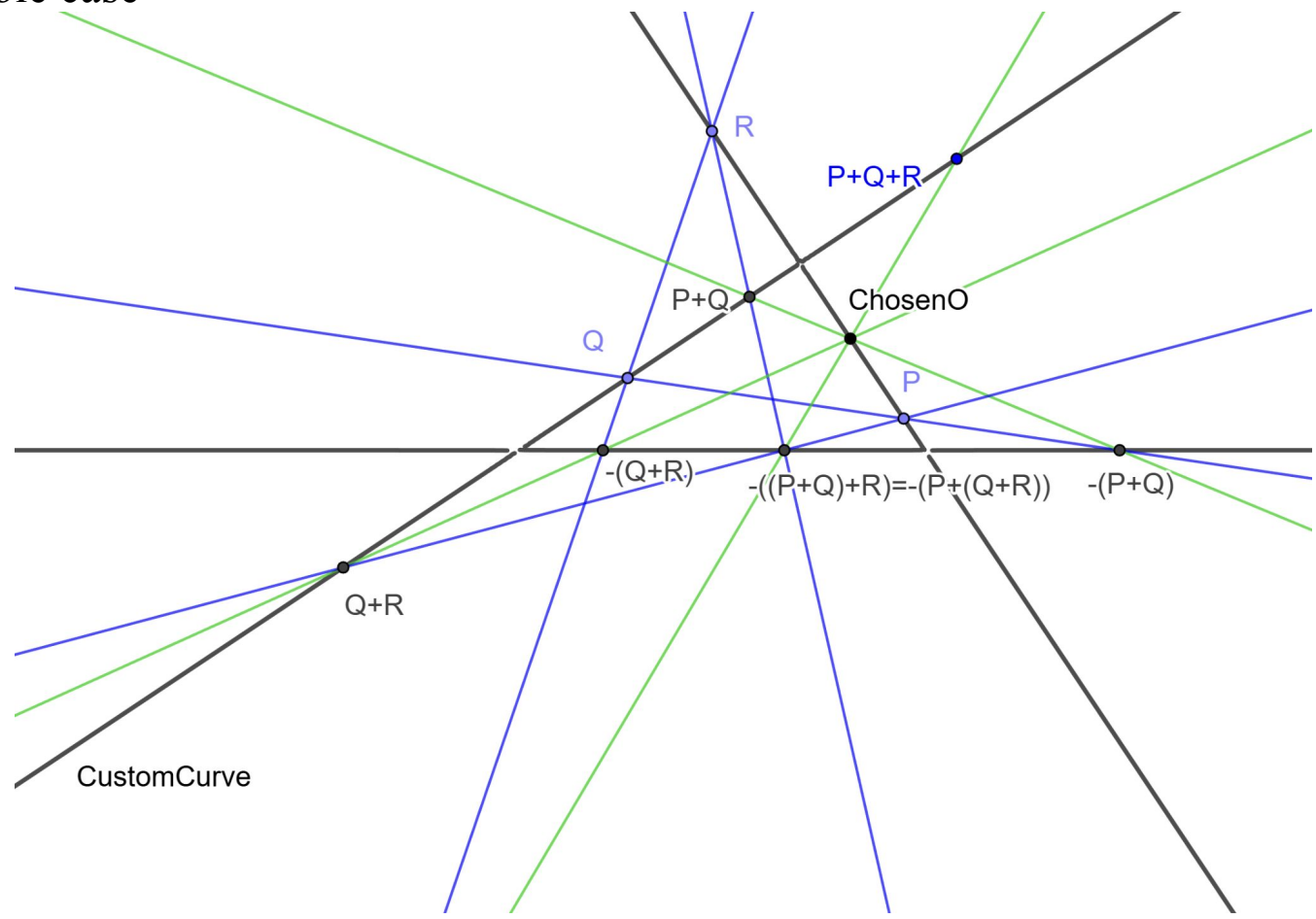
Partial semigroup

CustomCurve

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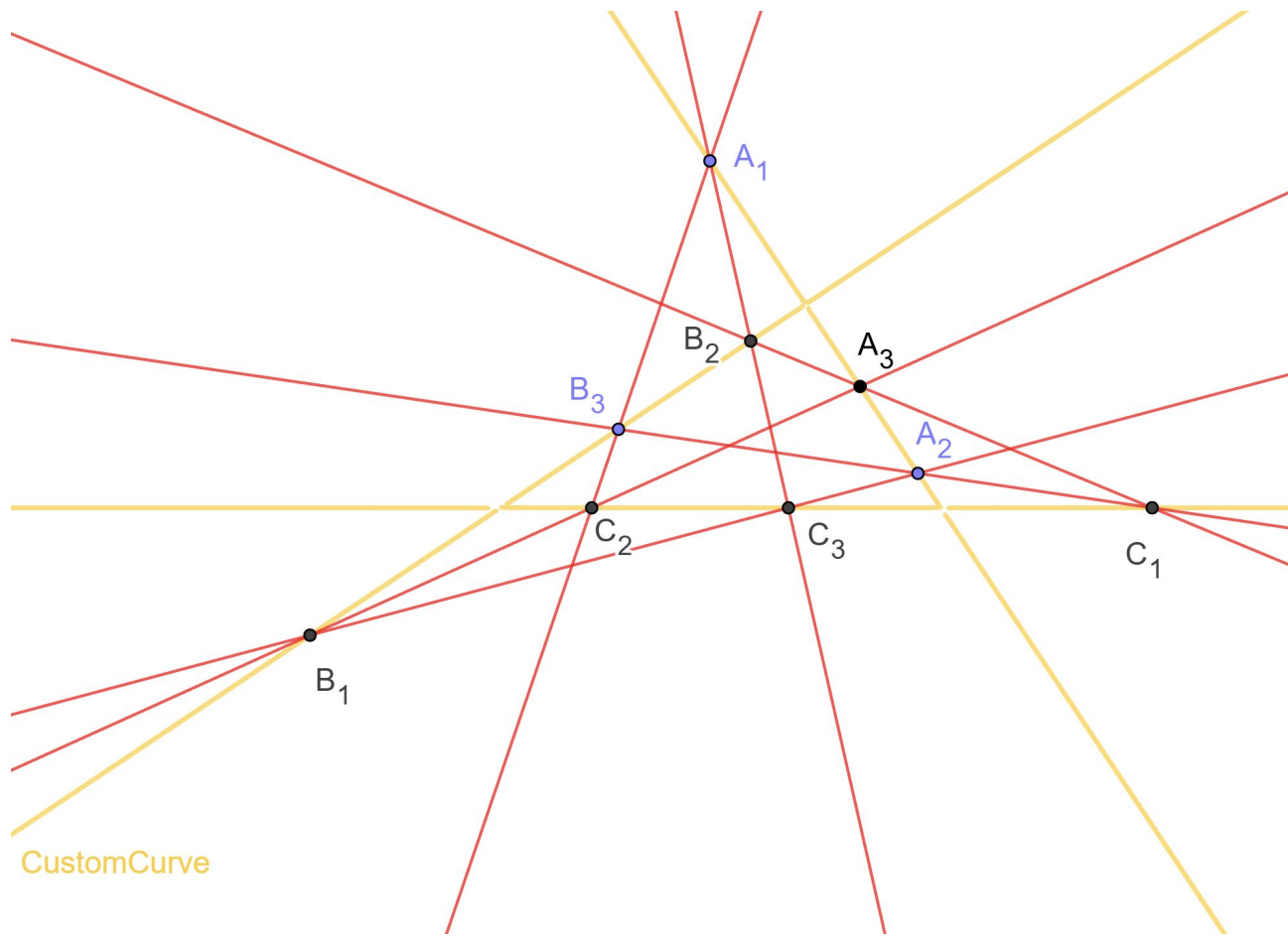


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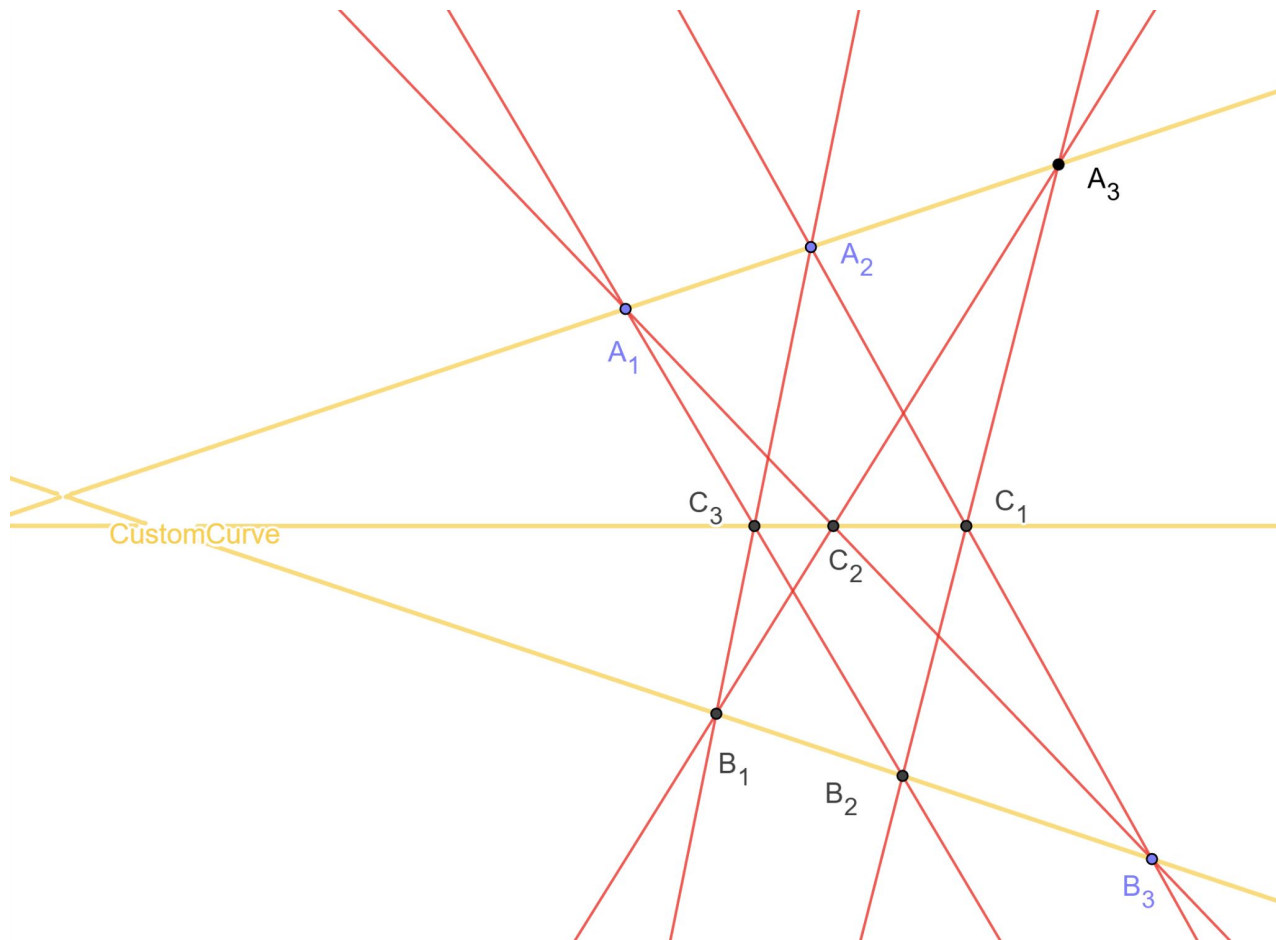
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Pappus Theorem

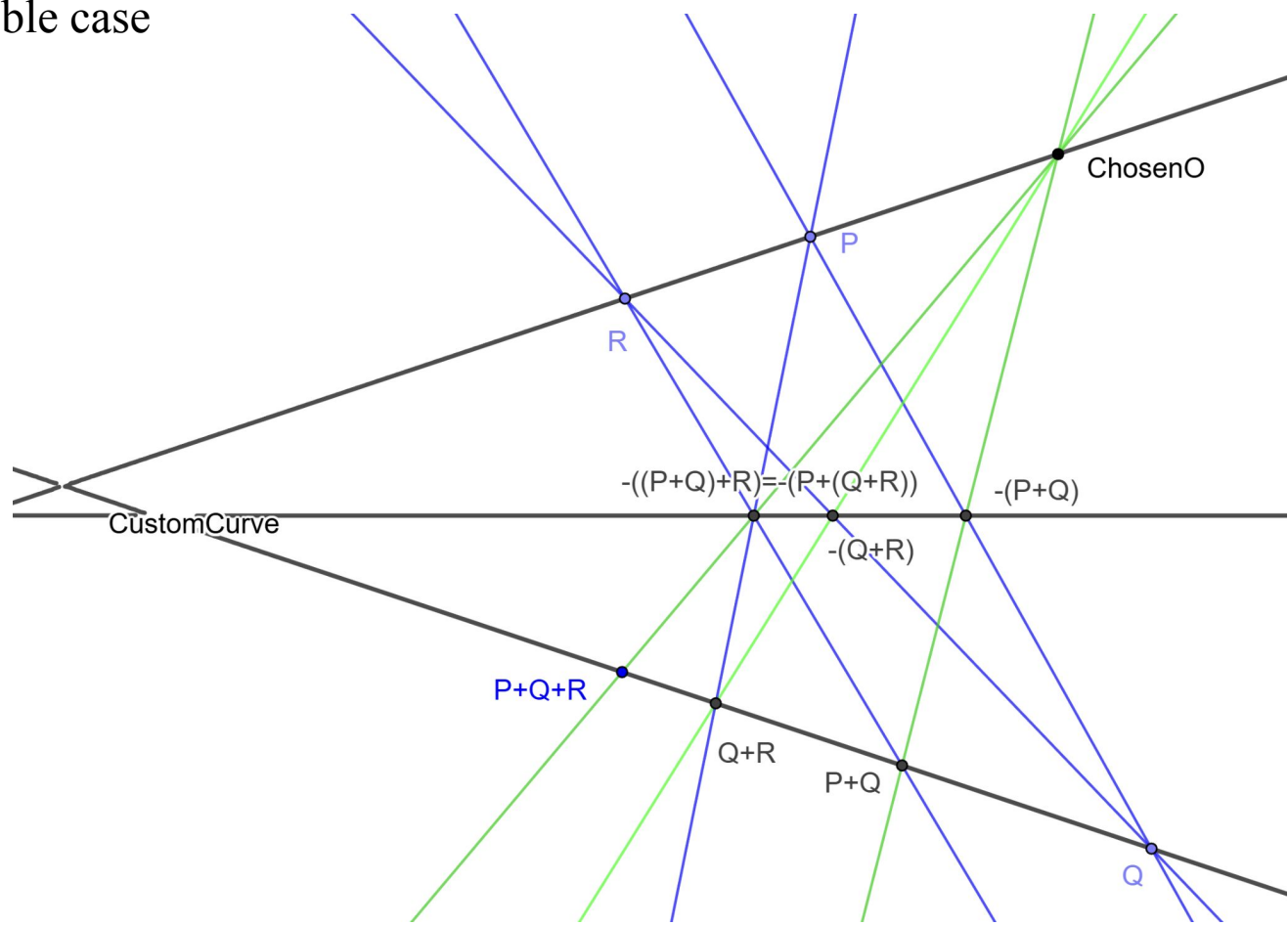
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Pappus Theorem

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